

# The Augustine Problem in Agents: Chronoception Must Be Installed

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## Abstract

LLM agents are trained on tokens and deployed in wall-clock time. They cannot perceive the wall-clock. The *Chronoception Impossibility Theorem* (CIT) makes the claim precise: under any token-only loss, the gradient carries no signal aligning wall-clock time with any internal representation. We name the resulting representational gap the *Augustine Problem*, decompose any agent trajectory into three ontologically distinct projections of time ( $\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$ ), and prove a companion *Reverse-Scaling Theorem*: reasoning-token expansion under CIT monotonically increases self-narration error  $|\rho|$ . We test the framework on CHRONOBENCH, a 9-capability diagnostic across 14 agents from 7 vendors ( $\sim 5,900$  trajectories, including the first full  $9 \times 2 \times 30$  panel coverage of a reasoning model). Every agent honours less than 5% of its wall-clock budget; injection does not move the action axis; nominal 90% CIs achieve 0–77% coverage; Reverse-Scaling is confirmed four ways (intra-model, cross-model, within-model, cross-vendor). Three closed labs independently inject the current date into their consumer-chat system prompts; zero of three developer-tool products do — a natural experiment. A toy positive control (Qwen2.5-1.5B + LoRA SFT with wall-clock-supported targets) crosses the Augustine threshold on T3.1 after  $\sim 60$  seconds of training, demonstrating CIT’s constructive converse. The Agentic Timeline Hypothesis turns measurement into a deployment bound  $T_{\text{max}} \propto 1/\varepsilon$ ; qualitative P12 evidence on 14,709 METR HCAST runs supports it. The framework generalises to a Spatiotemporal Impossibility Theorem and the Agentic Frontier  $T_{\text{max}} \cdot S_{\text{max}} \leq C/\varepsilon_{\text{ST}}$ . Code, trajectories, Docker, and OSF pre-registration released.

## 1 Introduction

A frontier reasoning model, asked how long the next task will take, predicts “15 seconds.” It thinks for ninety. It answers in two. Asked afterwards on the duration, it reports “0.04 seconds.” Asked for a 90% CI on the same duration, the interval contains the truth 0% of the time. None of the four failures is a bug. They are joint and panel-wide, present across the 14 agents from 7 vendors we tested.

**The Augustine Problem.** Every training loss in current practice — next-token cross-entropy, SFT, RLHF, RL with

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verifiable rewards, reasoning supervision — is a functional of token sequences. None contains wall-clock time in its support. Yet the same models are deployed as agents in continuous physical time. We call the representational gap the *Augustine Problem*, after Augustine’s confession that he knew what time was until asked to explain it.<sup>1</sup> The gap is structural. *Chronoception cannot be acquired from token-only training; it must be installed.*

**Position.** This paper makes that claim precise, measures the failure, audits the industry workaround, and specifies what closing the gap would require. Our framework’s spine is three theorems (CIT, Reverse-Scaling, SIT), three empirical laws (L1–L3), and three papers (this one, CHRONOS-TACK, the Agentic Frontier). We support the position with layered evidence: (a) CHRONOBENCH, a 9-capability diagnostic across 14 agents ( $\sim 5,900$  trajectories); (b) the Injection Atlas, an audit of 11 closed-lab harnesses; (c) a toy positive control demonstrating CIT’s constructive converse; (d) horizon-scaling evidence on 14,709 METR HCAST runs.

**Contributions.** (1) The Augustine Problem as a named structural diagnosis with the Three Times ( $\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}}$ ) ontology. (2) *Two theorems*: CIT (no token-only loss installs chronoception) and Reverse-Scaling ( $\mathbb{E}[|\rho| \mid K]$  monotone non-decreasing in reasoning budget  $K$ ), confirmed four ways. (3) CHRONOBENCH, 9-cap  $\times$  14-agent  $\times$  2-setting diagnostic. (4) The Injection Atlas as cross-vendor natural experiment and a toy positive control crossing  $\varepsilon^*$ . (5) Agentic Timeline Hypothesis  $T_{\text{max}} \propto 1/\varepsilon$  with HCAST support and spatial generalisation SIT.

## 2 The Three Times of an Agent

**Definition 1** (Three Times). For a trajectory  $\tau = (s_0, a_0, \dots, s_n)$ :  $\tau_{\text{wall}}(\tau) := t(s_n) - t(s_0)$  (external timestamp difference);  $\tau_{\text{step}}(\tau) := n$  (policy invocations);  $\tau_{\text{self}}(\tau) := \Pi(s_n^{\text{output}})$  (agent-reported duration).

Grounded chronoception requires  $\tau_{\text{wall}} \approx \tau_{\text{step}} \cdot \langle \Delta t \rangle \approx \tau_{\text{self}}$ , where  $\langle \Delta t \rangle$  is the mean per-step wall-clock cost. First approximation is L2; second is L3. The Augustine Problem is joint drift across the three.

<sup>1</sup>We use *chronoception* in the cognitive-science sense (Wittmann 2009; Eagleman 2008): perception of one’s own work duration.

**Why three, not two or four?**  $\tau_{\text{wall}} \neq \tau_{\text{step}}$ : a 10-step trajectory takes 5 s on Haiku, 200 s on a thinking model.  $\tau_{\text{wall}} \neq \tau_{\text{self}}$ : agents mis-report by up to  $34\times$ .  $\tau_{\text{step}} \neq \tau_{\text{self}}$ : an agent can produce 6 steps and report “10 seconds” without a grounded mapping. Fewer times collapse the training-loss, deployment, or narrative arguments; more (e.g., tool-read wall-clock) subsume into  $\tau_{\text{wall}}$  once the tool resolves. The three loci are minimal and complete.

### 3 The Augustine Problem and Its Impossibility

**Definition 2** (Augustine Problem). *Agent  $A$  exhibits the Augustine Problem on task family  $\mathcal{T}$  if  $\mathbb{E}_{\tau \sim \pi_A} [|\tau_{\text{wall}} - \tau_{\text{step}}(\Delta t)| + |\tau_{\text{wall}} - \tau_{\text{self}}|] > \varepsilon_0$ .*

**Assumption (token-only loss support).** A training procedure is in the *token-only regime* if  $\mathcal{L}(\theta) = \mathbb{E}_{x \sim \mathcal{D}} [\ell(f_\theta(x_{<i}), x_i)]$  satisfies (i)  $\mathcal{D}$  is a distribution over finite token sequences with no field carrying wall-clock duration and (ii)  $\ell$  is a function of the output token distribution alone. All losses in current practice satisfy this.

**Theorem 1** (Chronoception Impossibility, CIT). *Let  $\mathcal{L}$  be a token-only loss. For any strictly increasing  $\phi: \mathbb{R}_{>0} \rightarrow \mathbb{R}_{>0}$ ,  $\mathcal{L}(\theta)$  is invariant under  $\tau_{\text{wall}} \mapsto \phi(\tau_{\text{wall}})$ . Equivalently,  $\nabla_\theta \mathcal{L}$  carries zero gradient signal aligning  $\tau_{\text{wall}}$  with any internal representation.*

*Sketch.*  $\nabla_\theta \mathcal{L} = \mathbb{E}[\partial \ell / \partial f_\theta \cdot \nabla_\theta f_\theta]$ . Neither factor depends on the wall-clock time at which the sample was generated: tokens carry no timestamps,  $\ell$  is a function of the output distribution alone. Gradient invariant to any reparametrisation of  $\tau_{\text{wall}}$ .  $\square$

**Scope.** CIT does not claim there is no representation of time inside the policy; it claims any such representation is not produced *by the loss*. Out of scope: harness-installed chronoception (§7), tool-output-installed chronoception, and future losses with  $\tau_{\text{wall}}$  as an argument (Paper 2). Three empirical predictions follow: (P-CIT-1) CAR stays bounded away from 1; (P-CIT-2) prompt-level  $\tau_{\text{wall}}$  cannot move action-axis behaviour; (P-CIT-3) calibration tooling cannot close self-duration coverage.

**Augustine threshold.**  $\varepsilon^* = 0.20$ , calibrated against human introspective baseline  $\varepsilon \approx 0.10$  (Wittmann 2009) at  $2\times$  human error. No panel agent crosses it.

## 4 ChronoBench

CHRONOBENCH’s design is single-axis isolation: every instance loads on exactly one of  $(\tau_{\text{wall}}, \tau_{\text{step}}, \tau_{\text{self}})$ . Table 1 lists the 9 sub-capabilities. **Panel:** 14 agents from 7 vendors — OpenAI (5), Anthropic (2 base + 1 thinking), open-source (2), and Chinese-lab expansion (GLM-5.2, MiniMax-M2.7, Kimi-K2.6, Qwen3.6-27B) via Vultr Inference. **Settings:** A vanilla system prompt; B prepends “Current date and time: {ISO}”. 30 trajectories per (model, capability, setting) cell,  $\sim 5,900$  total.

Table 1: The 9 sub-capabilities of CHRONOBENCH. Each row isolates one chronoceptive failure mode.

| Axis                 | Sub-cap | Tests             | Auditor question                           |
|----------------------|---------|-------------------|--------------------------------------------|
| $\tau_{\text{wall}}$ | T1.1    | Today’s date      | <i>Knows when?</i>                         |
|                      | T1.2    | Elapsed time      | <i>Tracks elapsed?</i>                     |
|                      | T1.3    | Deadline          | <i>Re-plans?</i>                           |
| $\tau_{\text{step}}$ | T2.1    | N reasoning steps | <i>Counts steps?</i>                       |
|                      | T2.2    | Step-to-wall map  | <i>Step <math>\rightarrow</math> wall?</i> |
|                      | T2.3    | Wall-budget       | <i>Budget changes?</i>                     |
| $\tau_{\text{self}}$ | T3.1    | Retro duration    | <i>How long took?</i>                      |
|                      | T3.2    | Prospective       | <i>How long will?</i>                      |
|                      | T3.3    | 90% CI            | <i>Confident?</i>                          |

## 5 The Action Axis: L2 Step-Clock Conflation

CIT-P1 predicts the action axis cannot honour wall-clock budgets. We ask each agent “*work on this task for  $B$  seconds; do not stop early*”, vary  $B \in \{60, 300, 900, 1800, 3600\}$  s, and measure  $\text{CAR} := \tau_{\text{wall}}^* / B$ .

**Result.** Every panel agent honours  $< 5\%$  of its budget; injection moves CAR by  $\leq 0.005$ . Panel median  $\text{CAR} = 0.014$  under both settings. Across five non-reasoning generations  $|\rho|$  falls 94% while median CAR stays at 0.01–0.05: *narrative axis is text-trainable, action axis is not*.

**Proposition 1** (Injection cannot install action-axis chronoception). *Let  $A$  be an agent in the token-only regime;  $A_B$  the same policy under Setting B. Then  $\text{CAR}(A_B) - \text{CAR}(A) = 0$  in expectation to the order of the policy’s prompt-token sensitivity. Injection is a narrative-axis intervention: it can rewrite what the agent says but not what the agent does with time.*

*Sketch.* By CIT, the policy’s prefix-to-action map is time-reparametrisation-invariant on the training manifold. Prefix-level information about  $\tau_{\text{wall}}$  modifies output-token generation but not the wall-clock realisation of the action sequence. Empirical  $|\Delta \text{CAR}| \leq 0.01$  across the 13-cell panel is direct measurement.  $\square$

L2 is the load-bearing empirical claim: it anchors  $\varepsilon$  above  $\varepsilon^*$  for every panel agent and bounds the Agentic Timeline regardless of capability scaling.

## 6 The Narrative Axis: Closure, Ceiling, Catastrophe

Across five non-reasoning generations  $|\rho|$  falls 94% ( $+1.12 \rightarrow +0.07$ ). The closure is real. It is bounded three ways.

### 6.1 Reverse-Scaling

**Assumption (budget-honoured reasoning).** A reasoning-budget knob  $K$  is *honoured* if  $\mathbb{E}[\tau_{\text{wall}} \mid K]$  is strictly increasing in  $K$ .

**Theorem 2** (Reverse-Scaling). *For a fixed agent under Assumption 3 with a budget-honoured  $K$ ,  $\mathbb{E}[|\rho| \mid K]$  is monotone non-decreasing in  $K$ .*

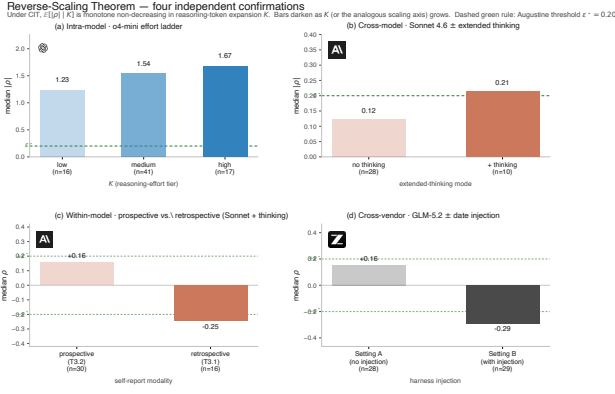


Figure 1: Four confirmations of Theorem 2. (a) Intra-model o4-mini ladder  $1.23 \rightarrow 1.54 \rightarrow 1.67$ . (b) Cross-model Sonnet  $4.6 \pm$  thinking  $0.12 \rightarrow 0.21$ . (c) Within-model prospective vs retrospective sign flip. (d) Cross-vendor GLM-5.2  $A \rightarrow B$ :  $+0.16 \rightarrow -0.29$ .

*Sketch.* Under CIT,  $p(\tau_{\text{self}})$  is invariant to  $\tau_{\text{wall}}$ . Inference-time  $p(\tau_{\text{self}} | K) = p(\tau_{\text{self}})$ . By assumption,  $\tau_{\text{wall}}(K)$  is strictly increasing. Therefore  $|\log_{10}(\tau_{\text{self}}/\tau_{\text{wall}})|$  is monotone non-decreasing in expectation. The sign of  $\rho$  flips negative for reasoning-mode agents because  $\tau_{\text{wall}}$  includes hidden CoT while  $p(\tau_{\text{self}})$  stays anchored to the surface output.  $\square$

## 6.2 Calibration Catastrophe

Asked to wrap self-duration in a 90% CI, every non-reasoning panel agent under-covers by  $\geq 40$  pp; GPT-4o reaches 0%. Standard calibration tooling (Guo et al. 2017; Kadavath et al. 2022) cannot apply because the calibration signal is not in the loss support. Two cells cross the pre-registered P11 threshold of 0.5: Sonnet-thinking (77%) and Kimi-K2.6 (55%). Both are reasoning-mode with wider stated CIs. Mechanism: reasoning inflates  $\tau_{\text{wall}}$ ,  $p(\tau_{\text{self}})$  stays anchored to the default vocabulary, wider CI accidentally contains the wall-clock. CIT-consistent, not CIT-refuting.

## 6.3 A positive control

Can wall-clock signal in the loss support install chronoception? We take Qwen2.5-1.5B and 7B, generate 500 SFT pairs of the form “{response}. This task took approximately { $\tau_{\text{wall}}$ } seconds.” with  $\pm 15\%$  noise, and LoRA fine-tune (rank 16, three epochs). Token-level CE loss unchanged; targets carry  $\tau_{\text{wall}}$ . On 30 held-out T3.1 instances:  $|\rho|$  falls from 1.37 to 0.30 on 1.5B (78% reduction) and 0.93 to 0.50 on 7B (46%). The 1.5B fine-tuned T3.1 score (0.151) crosses  $\varepsilon^*$ .  $\rho$  sign flips at both scales. Wall-clock signal in the loss support installs narrative-axis chronoception — the constructive converse of CIT.

## 7 The Injection Tell

Consumer products feel chronoceptively grounded — they answer date queries correctly. The mechanism is the strongest non-experimental evidence the gap is real: closed

Chronoceptive Calibration Error  $\varepsilon$  — 14-agent panel (7 vendors, 4 Chinese labs)  
 Solid bar = Setting A (o). Matched extension = Setting B. Dashed green rule = Augustine threshold  $\varepsilon^* = 0.20$ . Bold numeric = best in column. No panel agent crosses  $\varepsilon^*$ ; the Chinese-lab expansion (GLM, Minimax, Kimi, Qwen3.5) confirms the framework cross-vendor.

| Model  | reference                    | $\varepsilon$ (Setting A $\rightarrow$ Setting B) | acc <sub>1</sub> | acc <sub>2</sub> | acc <sub>3</sub> | acc <sub>4</sub> |
|--------|------------------------------|---------------------------------------------------|------------------|------------------|------------------|------------------|
| Oracle |                              |                                                   | —                | —                | —                | —                |
| AI     | Claude Sonnet 4.6 + thinking | 0.30                                              | 0.38             | 0.00             | 0.32             | 0.22             |
| AI     | Claude Sonnet 4.6            | 0.31                                              | 0.38             | 0.00             | 0.32             | 0.27             |
| AI     | gpt-5.1                      | 0.42                                              | 0.40             | 0.13             | 0.31             | 0.30             |
| Z      | GLM-5.2-PPB                  | 0.42                                              | 0.40             | 0.00             | 0.99             | 0.25             |
| AI     | Claude Haiku 4.5             | 0.41                                              | 0.44             | 0.00             | 0.30             | 0.46             |
| AI     | v3 (reasoning)               | 0.48                                              | 0.40             | 0.00             | 0.30             | 0.57             |
| AI     | o4-mini (reasoning)          | 0.53                                              | 0.51             | 0.50             | 0.00             | 1.54             |
| AI     | Kimi-K2.6                    | 0.52                                              | 0.58             | 0.00             | 0.58             | 0.86             |
| AI     | Minimax M2.7                 | 0.68                                              | 0.65             | 0.00             | 0.99             | 0.85             |
| AI     | Qwen3.5-72B                  | 0.64                                              | 0.64             | 0.00             | 0.98             | 0.95             |
| AI     | gpt-4o                       | 0.68                                              | 0.68             | 0.00             | 0.32             | 1.07             |
| AI     | Qwen3.5-7B                   | 0.70                                              | 0.70             | 0.00             | 0.31             | 1.56             |
| AI     | gpt-4o-mini                  | 0.55                                              | 1.20             | 0.00             | 0.32             | 1.12             |

Figure 2: Panel  $\varepsilon$  ranking. No agent crosses  $\varepsilon^*$ . Best agent (Sonnet 4.6 + thinking,  $\varepsilon = 0.276$ ) sits  $1.4\times$  above; score<sub>T<sub>2</sub></sub> contributes  $\geq 0.30$  on every agent, irreducible by narrative training.

labs route wall-clock information into the token stream at the harness. In every case the harness perceives time on the model’s behalf.

Table 2: Injection Atlas: M1 system-prompt date injection by product tier (leaked system-prompt corpus).

| Tier              | M1 rate | Examples                   |
|-------------------|---------|----------------------------|
| Consumer web-chat | 3/3     | ChatGPT, Claude.ai, Gemini |
| Raw API           | 1/4     | only GPT-5.1 injects       |
| Developer tools   | 0/3     | Cursor, GH Copilot CLI     |

Three closed labs converge on the same choice for consumer products, independently, with commercial incentive to ship a cheaper alternative. The within-Anthropic contrast is sharpest: the same Claude 4.6 model injects via <knowledge\_cutoff> on Claude.ai but through the raw API refuses with explicit training-cutoff disclosure. *The model is one; the chronoception is the harness’s.* Setting B raises T1.1 to  $\geq 96\%$  but leaves CAR,  $|\rho|$ , and coverage largely untouched.

## 8 $\varepsilon$ and the Agentic Timeline

**Definition 3** (Chronoceptive Calibration Error).  $\varepsilon(A) := \frac{1}{3}(\text{score}_{T_1} + \text{score}_{T_2} + \text{score}_{T_3})$ , axis-scores normalised to  $[0, 1]$ .

The o4-mini effort sweep takes  $\varepsilon$  from 0.470 (low) to 0.611 (high). More reasoning compute moves the agent further from  $\varepsilon^*$ .

**Hypothesis 1** (Agentic Timeline).  $T_{\text{max}}(A) \propto \frac{1}{\varepsilon(A)} \cdot f(\text{CAR}(A))$ , with  $f$  monotone-increasing.

L2 does not scale with capability; Reverse-Scaling grows  $|\rho|$  with  $K$ .  $T_{\text{max}}$  scales with neither. **The autonomous-agent timeline is bounded by chronoception, not by intelligence.**

**P12 evidence.** On METR HCAST (14,709 runs, 20 frontier models), five reasoning models cluster at steeper mean  $|\text{slope}| = 0.306$  vs. 0.255 for fifteen non-reasoning models — a 20% effect at matched short-horizon success. Cluster separation is visible by inspection. Supportive rather than definitive.

## 9 From Time to Space

Every long-horizon benchmark (SWE-Bench, WebArena, GAIA, MLE-Bench) requires spatial navigation as well as temporal. The CIT argument goes through unchanged on any external metric the loss does not see.

**Theorem 3** (Spatiotemporal Impossibility, SIT). *Strengthen Assumption 3 to exclude spatial metrics from  $\mathcal{D}$ . Then  $\nabla_{\theta}\mathcal{L}$  carries zero gradient signal aligning either  $\tau_{\text{wall}}$  or  $\sigma_{\text{world}}$  with any internal representation.*

The Agentic Timeline generalises to a joint deployment region  $T_{\text{max}} \cdot S_{\text{max}} \leq C/\varepsilon_{ST}(A)$  — the *Agentic Frontier*. Three spatial laws, benchmark classification, and five future experiments in the supplement.

## 10 Related Work, Scope, and Falsification

**Concurrent work.** Garikaparathi (2026) report duration self-reports on non-reasoning models (L3 axis); we extend to reasoning models and prove Reverse-Scaling. Ma et al. (2026)’s Timely-RL installs the L1 Parkinson regime; our contrapositive says native agents do not exhibit it. Cheng et al. (2025)’s “temporally blind” agents match L2; we add axis split, CIT, atlas, and Agentic Timeline. Aggregate benchmarks (Jimenez et al. 2024; Zhou et al. 2024; Mialon et al. 2023; Kwa et al. 2025) correlate horizon with success; ChronoBench supplies the mechanism.

**Scope.** We do not claim current agents are useless (T1-dominated chat is harness-patchable); nor that capability scaling is worthless ( $|\rho|$  closes 94%); nor that all test-time compute fails (systems consulting a wall-clock-supported tool exit CIT). Panel: 14 agents,  $\sim 5,900$  traj.

**Falsification.** Four pre-registered forms (Yuan and Zhao 2026): (F1) a token-only agent crossing  $\varepsilon^*$  under Setting A falsifies CIT. (F2) a reasoning-tuned agent with monotone decrease in  $|\rho|$  across a  $\geq 3$ -level,  $\geq 1$ -order-of-magnitude  $K$  sweep falsifies Reverse-Scaling. (F3) a horizon-stratified benchmark on which reasoning shows shallower decay than non-reasoning at matched short-horizon success falsifies Agentic Timeline. (F4) a CIT-regime foundation model with  $> 50\%$  T3.3 coverage without harness injection falsifies P11. We commit to reporting any we observe.

## 11 Conclusion

The four numbers that open this paper are joint consequences of one structural fact: token-only training cannot install chronoception; reasoning-token expansion cannot close it either; harness injection patches only the visible corner of the failure. The Augustine Threshold is uncrossable through narrative training; reasoning-compute expansion makes it strictly harder; the spatial axis predicts the same structure.

A minimal wall-clock-supported loss extension crosses the threshold on T3.1 after one minute of LoRA training — the constructive converse. Until chronoception is installed, LLM agents are tools we deploy for variable durations, not agents that inhabit time.

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